OPENSOURCESDRLAB DOCUMENTATION

B210-Kintex-7 Quick Start Guide

Product	B210-Kintex-7
Document	OSSDRLAB-B210-Kintex-7-QSG
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1. Overview

The B210-Kintex-7 is a USRP B210-compatible software-defined radio. It keeps the AD9361 RF transceiver and the frequency-divided RF front end of the original USRP B210, replaces the Spartan-6 FPGA with a Xilinx Kintex-7 XC7K325T, connects to the host through USB 3.0 Type-C, and adds an onboard GPS module for PPS timing. The GPSDO slot of the original board is removed, which brings the size down to 70 × 97 × 11.5 mm.

The board works with the standard Ettus Research UHD driver and UHD-based software. Because the FPGA is different, it needs its own FPGA image, `usrp_b210_fpga.bin`; this guide shows how to install it and check the board.

Differences in use compared with the original USRP B210:

- the stock `usrp_b210_fpga.bin` from Ettus Research must be replaced with the image for this board;
- a GPSDO module cannot be fitted;
- the PPS from the onboard GPS module has priority over the external PPS input: while the GPS module delivers PPS pulses, the external PPS input is not used.

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2. What you need

Item	Notes
B210-Kintex-7 board	
USB 3.0 Type-C cable	connect to a USB 3.0 (blue) port; USB 2.0 ports do not give enough bandwidth
Antennas or RF cables	SMA
GPS antenna (optional)	MMCX, for the onboard GPS module
PC	64-bit Windows 10/11 or Ubuntu Linux
UHD driver	from Ettus Research: https://files.ettus.com/binaries/uhd/latest_release/ (Windows installer), or the `uhd-host` package on Ubuntu
WinUSB driver (Windows only, if needed)	from Ettus Research: https://files.ettus.com/binaries/misc/erllc_uhd_winusb_driver.zip
`libusb-1.0.dll` (Windows only, if UHD reports it missing)	from the libusb project (64-bit Windows binaries)
FPGA image for this board	`B210-Kintex-7_FPGA-image_Vivado2024.1.zip` from the Releases page

3. Connectors and indicators

Marking	Connector	Function
TRX1, TRX2	SMA	transmit or receive port, channel 1 / 2
RX1, RX2	SMA	receive port, channel 1 / 2
USB	USB 3.0 Type-C	data and power from the host
PPS	MMCX	external PPS input
10M	MMCX	external 10 MHz reference input
GPS	MMCX	antenna input of the onboard GPS module
FPC connector	FPC	FPGA JTAG and spare GPIO

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LED	Meaning
PWER	power; red when the board is powered
STAS	status; blue after the FPGA image has been loaded
CLK	PPS; flashes red in step with the selected PPS source
USR	no function by default, free for user designs
TRX1, TRX2	red while transmitting, blue while receiving
RX1, RX2	blue while receiving

4. Set up on Windows

1. Download the UHD installer for your Windows version from Ettus Research (see section 2), right-click it and run it as administrator. Keep the default installation folder `C:\Program Files\UHD`.
2. Connect the board. In Device Manager it should appear as **Ettus Research LLC B200/B210** under **USRPs**. If it shows up as an unknown device, choose **Update driver**, point it to the driver folder in the UHD installation, or unzip `erllc_uhd_winusb_driver.zip` and point it to that folder.
3. Make sure the standard UHD images (including the FX3 firmware `usrp_b200_fw.hex`) are in `C:\Program Files\UHD\share\uhd\images\`. If the folder is empty, download them in a Command Prompt run as administrator (needs Python):

   ```
   cd "C:\Program Files\UHD\lib\uhd\utils"
   python uhd_images_downloader.py
   ```

4. Unzip `B210-Kintex-7_FPGA-image_Vivado2024.1.zip` and copy `usrp_b210_fpga.bin` over the file of the same name in `C:\Program Files\UHD\share\uhd\images\`.
5. Open a Command Prompt in `C:\Program Files\UHD\bin` and run:

   ```
   uhd_find_devices
   uhd_usrp_probe
   ```

   `uhd_find_devices` should list a **B210** with its serial number, and `uhd_usrp_probe` should print the full device tree (FPGA version, RX/TX channels, frequency and gain ranges).

> **Tip:** if `uhd_usrp_probe` says that `libusb-1.0.dll` is missing, copy `libusb-1.0.dll` (64-bit) into `C:\Program Files\UHD\bin` and run it again.

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> **Tip:** the error `RuntimeError: fx3 is in state 5` means the stock Ettus FPGA image was loaded. Repeat step 4. `uhd_usrp_probe` prints the path of the image it loads.
>
> **Tip:** some applications bring their own copy of UHD and of `usrp_b210_fpga.bin` (for example SDR# with a USRP plug-in). Replace the image in that program's folder too.

5. Set up on Ubuntu Linux

1. Install UHD:

   ```
   sudo apt update
   sudo apt install -y libuhd-dev uhd-host
   ```

2. Download the standard images and replace the FPGA image:

   ```
   sudo uhd_images_downloader
   uhd_find_devices        # prints the images folder if the image is wrong
   ```

   Copy `usrp_b210_fpga.bin` from the release zip over the file of the same name in the UHD images folder (usually `/usr/share/uhd/images/`). If you keep the images in your own folder, set `UHD_IMAGES_DIR` to that folder.

3. Allow normal users to access the board (only needed if UHD reports a permission error):

   ```
   sudo tee /etc/udev/rules.d/10-usrp.rules >/dev/null << 'EOF'
   SUBSYSTEM=="usb", ATTRS{idVendor}=="2500", ATTRS{idProduct}=="0020", MODE="0666"
   EOF
   sudo udevadm control --reload-rules && sudo udevadm trigger
   ```

4. Connect the board to a USB 3.0 port and check it with `uhd_find_devices` and `uhd_usrp_probe` as on Windows.

6. First signals

Each channel has two SMA ports. In UHD-based software, the channel is chosen with the subdevice (`subdev`) and the port with the antenna setting:

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Board port	Subdevice	Antenna
TRX1	`A:A`	`TX/RX`
RX1	`A:A`	`RX2`
TRX2	`A:B`	`TX/RX`
RX2	`A:B`	`RX2`

For example, the Gqrx device string for the RX2 port is `uhd,type=b200,subdev=A:B` with the antenna set to `RX2`.

Once `uhd_usrp_probe` passes, any UHD-based application can use the board: GNU Radio, SDRangel, Gqrx, Universal Radio Hacker and others. Two GNU Radio Companion examples are in `examples/gnuradio/` of the repository: a two-channel receiver and a TX-to-RX loop-back test.

Step-by-step instructions for these applications are in the user manual on our blog: https://blog.opensourcesdrlab.com/archives/kintex-7-usrp-b210-user-manual

7. FPGA source project

Path	Contents
`fpga/B210_Project_Firmwire.xpr`	Vivado 2024.1 project, target `xc7k325tffg676-2`, top module `b200`
`fpga/B210_Project_Firmwire.srcs/sources_1/imports/.../lib/`, `.../top/b200/`	Ettus Research USRP FPGA sources (UHD v4.9.0.0), with the board changes listed in `NOTICE.md`
`fpga/top/b200/`	board constraints `b210.xdc`, PPS selection logic, halfband filter netlists
`fpga/B210_Project_Firmwire.srcs/sources_1/ip/`	Xilinx IP configurations (clocking wizard, FIFOs)

To rebuild the image:

1. Open `fpga/B210_Project_Firmwire.xpr` in Vivado 2024.1. Vivado regenerates the IP output products the first time.
2. Run **Generate Bitstream**. The project writes `b200.bit` and `b200.bin` to `B210_Project_Firmwire.runs/impl_1/`.
3. Rename `b200.bin` to `usrp_b210_fpga.bin` and install it as in section 4 or 5.

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8. Support

- Email: support@opensourcesdrlab.com
- Forum: https://forum.opensourcesdrlab.com
- Discord: https://discord.gg/nnxY6DkmvT
- Blog and user manuals: https://blog.opensourcesdrlab.com

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